

AmritChidiya (अमृतचिडिया)

Apni Sone Ki Chidiya Ko Phir Se Udaan Do 🐦 • Multilingual AI Welfare & Scholarship Companion

1. What is AmritChidiya? (Explained in Plain English)

Imagine you are a student or a farmer in a rural village. The government has hundreds of scholarships and monetary welfare programs for you — but they are locked behind complicated websites in English, difficult application criteria, and fraudulent WhatsApp scams demanding money. Most citizens give up and miss out on lakhs of rupees in legitimate financial help.

AmritChidiya changes this completely. It is a caring, AI-powered digital companion that speaks and listens to citizens in 5 major Indian languages (Hindi, Hinglish, English, Marathi, Tamil). Citizens can simply talk to it using voice or text. The AI calculates exactly which schemes they qualify for, guides them step-by-step through the application, warns them against scams, and even prints an official receipt they can take to their local village cyber cafe or Jan Seva Kendra.

2. The 4 Engineering Pillars & Member USPs

The platform is divided across 4 specialized team engineering roles, each with a game-changing technical innovation (USP):

Member	Engineering Role	Standout Technical USP (Unique Selling Proposition)
Member 1	Frontend UI/UX & Citizen Access Lead	Phygital Last-Mile Bridge & Citizen CSC Dossier: Created the printable CSC Dossier (AC-CSC-XXXXXX) with NPCI bank checklist and live Yojana Calculator, bridging online discovery with physical village cyber cafes.
Member 2	Backend API & Speech Processing Lead	Low-Latency Vernacular Audio Pipeline: Built sub-second voice pipeline using Groq Whisper Large v3 with Indian regional phonetic vocabulary bias and real-time speech text sanitization.
Member 3	AI Agent, Scheme Logic & Cyber Safety Lead	Anti-Hallucination Match Engine & Cyber Shield: Engineered a 0% hallucination LangGraph state machine paired with an in-app Cyber Shield verifying authentic .gov.in domains against WhatsApp fraud.
Member 4	Database, Auth Security & DevOps Lead	Zero-Trust Hybrid Cloud with Bcrypt Auth: Architected an enterprise-grade dual DB (Supabase + auto-failover SQLite) with 12-round Bcrypt salted security and signed 24-hr JWT tokens.

3. Core Platform Capabilities

- Multilingual Voice & Text: High-definition conversational interface across 5 languages (Hindi, Hinglish, English, Marathi, Tamil).
- Hands-Free Talk Mode: Full-duplex speech recognition via Groq Whisper Large v3 and natural Microsoft Edge-TTS neural speech.
- Yojana Eligibility Calculator: Dynamic evaluation matrix checking State, Category, Income, and Profession to compute instant benefit amounts.
- Official Domain Shield & Scam Checker: URL verification engine checking links against official Indian government registries (.gov.in, .nic.in).
- Phygital CSC Dossier Receipt: Printable formal receipt with NPCI/DBT bank checklist and VLE operator sign-off box for village cyber cafes.
- Direct Official Portal Links: Verified hyperlinking straight to central and state portals (scholarships.gov.in, pmkisan.gov.in).
- Enterprise Security: Bcrypt salted hashing (12 rounds), 24-hr signed JWT tokens, SlowAPI request throttling, and security headers.

4. Technology Stack

Layer	Technologies & Frameworks
Frontend	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Lucide Icons, ReactMarkdown
Backend API	FastAPI (Python 3.14), LangGraph, LangChain, Groq LLM (qwen/qwen3.8-27b)
Speech & Voice	Groq Whisper Large v3 (STT), Microsoft Edge-TTS (Neural Voice Synthesis)
Database & Auth	Supabase Cloud PostgreSQL + Local SQLite3 Failover, Bcrypt (12 rounds), Python-JOSE (JWT HS256)
Security & Ops	SlowAPI (Rate Limiting), Starlette Security Headers, CORS Origin Regex for Vercel/Render

5. Quick Local Setup

- Backend: `cd backend && python -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt && python main.py` (Runs on `http://127.0.0.1:8000`)
- Frontend: `cd frontend && npm install && npm run dev` (Runs on `http://localhost:3000`)